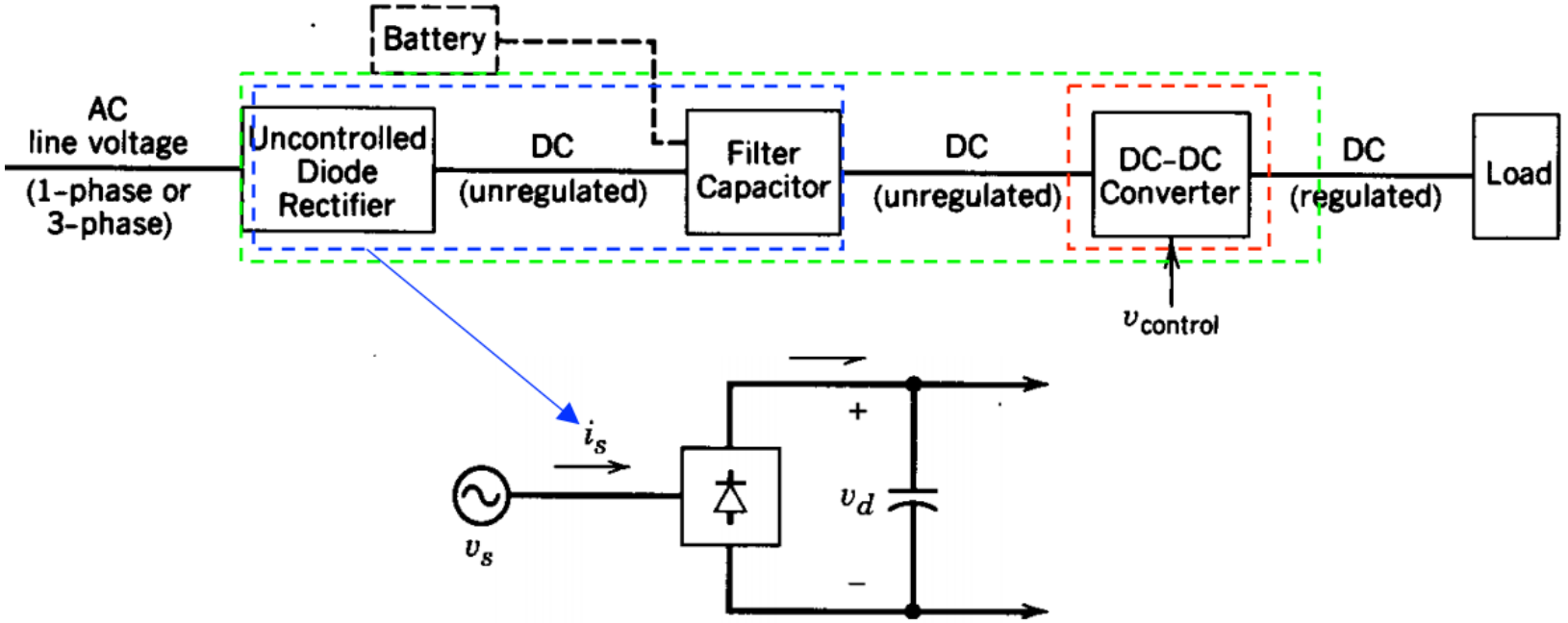
Chapter 4

DC-DC Switch-Mode Converters

Instructor: Le Hoa Nguyen, Ph.D.

DC-DC Converter System

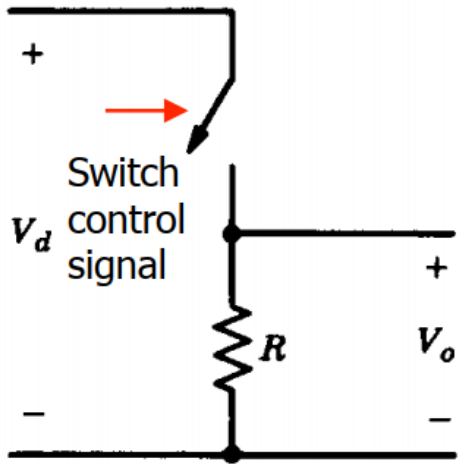


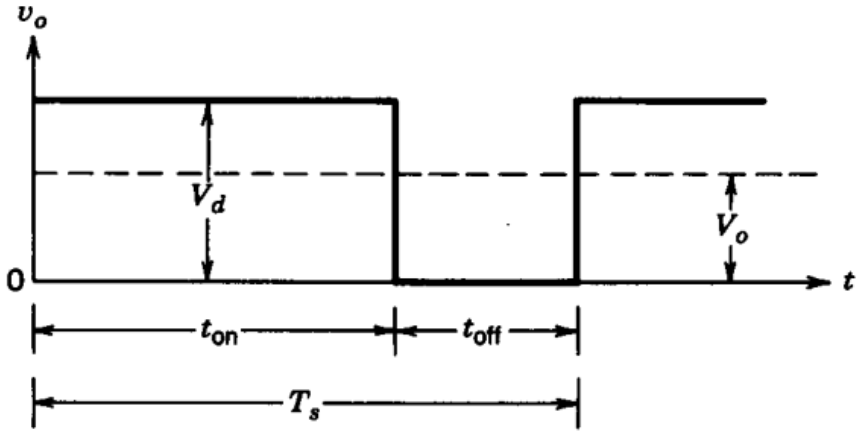
Types of DC-DC converter

- Step-down (buck) converter
- Step-up (boost) converter
- Step-down/step-up (buck-boost) converter
- Cúk converter
- Full-bridge converter

Control of DC-DC converter

Objective: To keep the output voltage (average value) to equal the desired level even though the input voltage and the output load may fluctuate



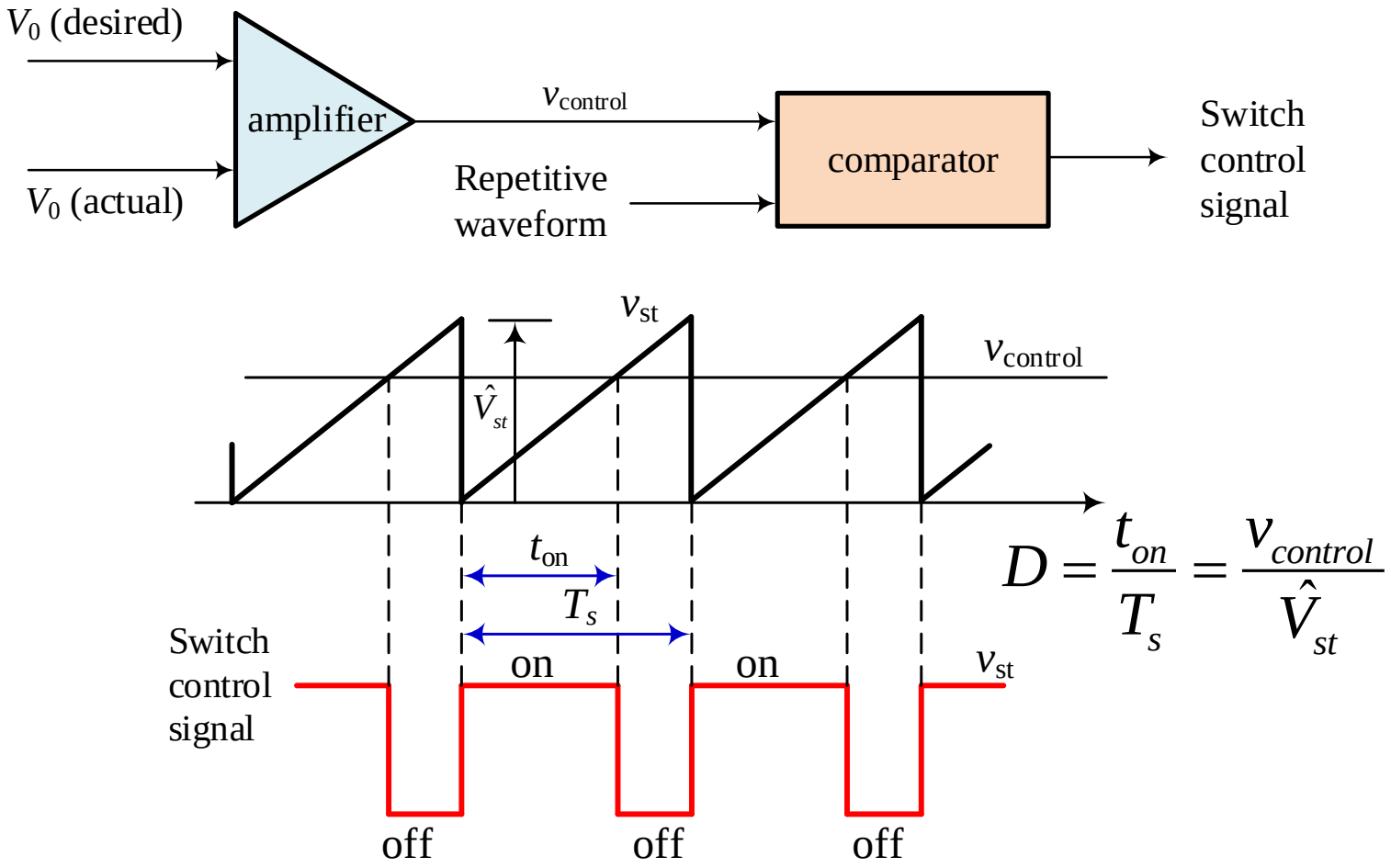


$$V_0 = \frac{1}{T_s}\int_0^{T_s} v_0(t)dt = \frac{1}{T_s}\left(t_{on}.V_d + t_{off}.0\right) = \frac{t_{on}}{T_s}V_d$$

Method:

- Keep frequency (hence T_s) to be constant and adjust the on duration of the switch (pulse-width modulation – PWM)
- Adjust both the switching frequency and the on duration of the switch

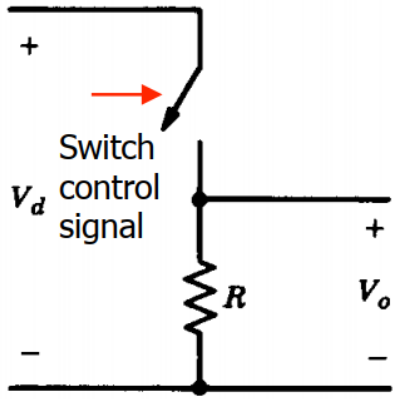
Control of DC-DC converter

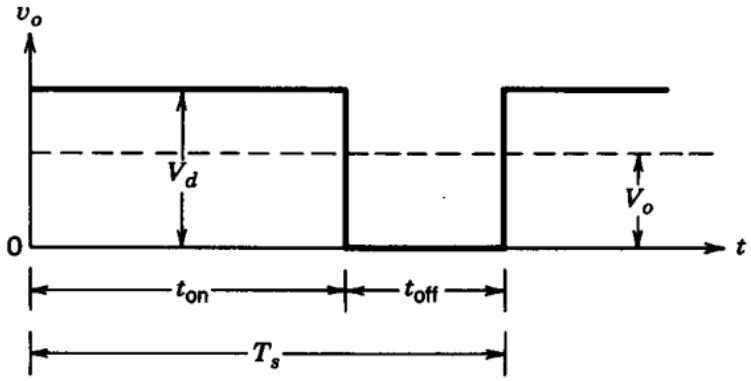


- The frequency of repetitive waveform establishes the switching frequency (few kHz to few hundred kHz)

Step-down (Buck) Converter

- A step-down converter produces a lower average output voltage than the dc input voltage
- Applications: Regulated power supply, dc motor speed control



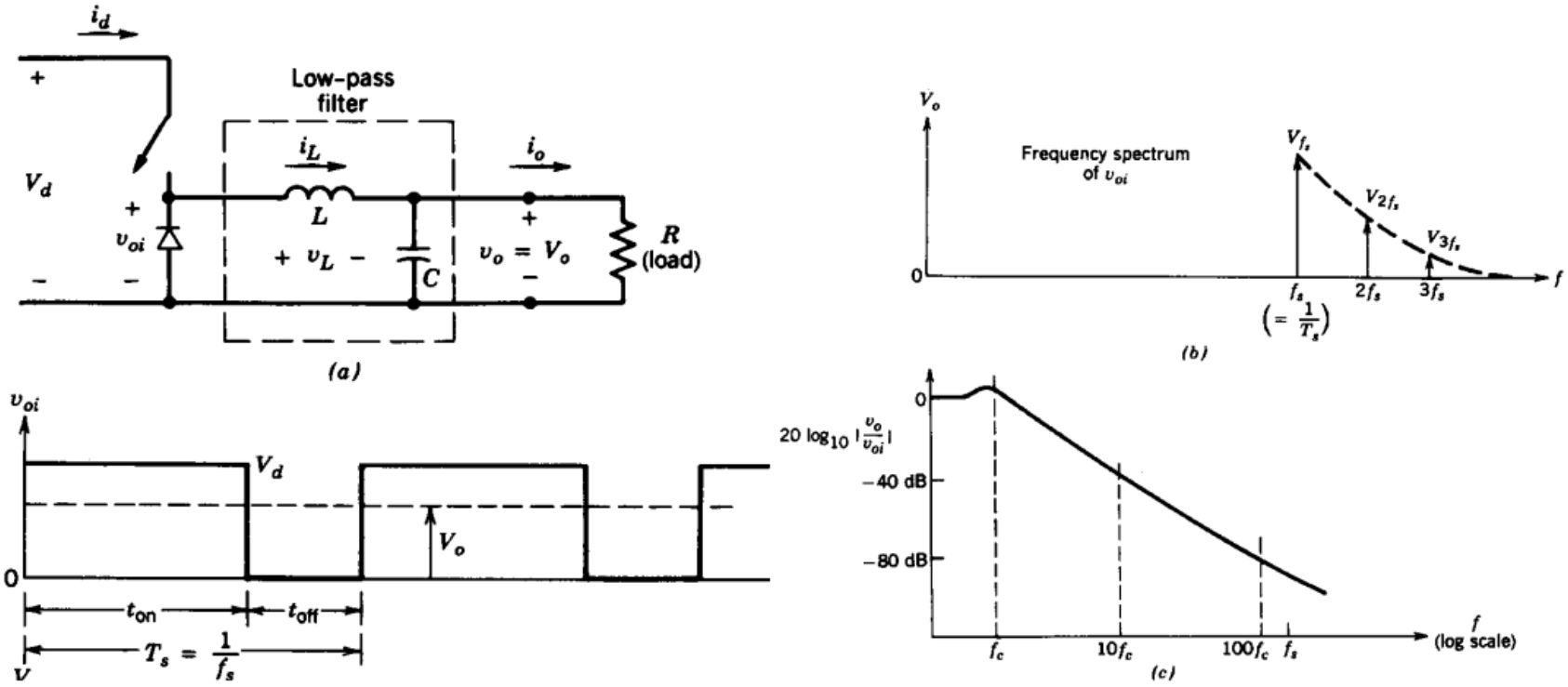


$$V_0 = \frac{1}{T_s}\int_0^{T_s} v_0(t)dt = \frac{1}{T_s}\left(t_{on}.V_d + t_{off}.0\right) = \frac{t_{on}}{T_s}V_d = DV_d \qquad V_0 = \frac{V_d}{\hat{V}_{st}}v_{control} = kv_{control}$$

Feature of this circuit

- The switch may be destroyed if load is inductive (the switch absorbs inductive energy)
- The output voltage fluctuates

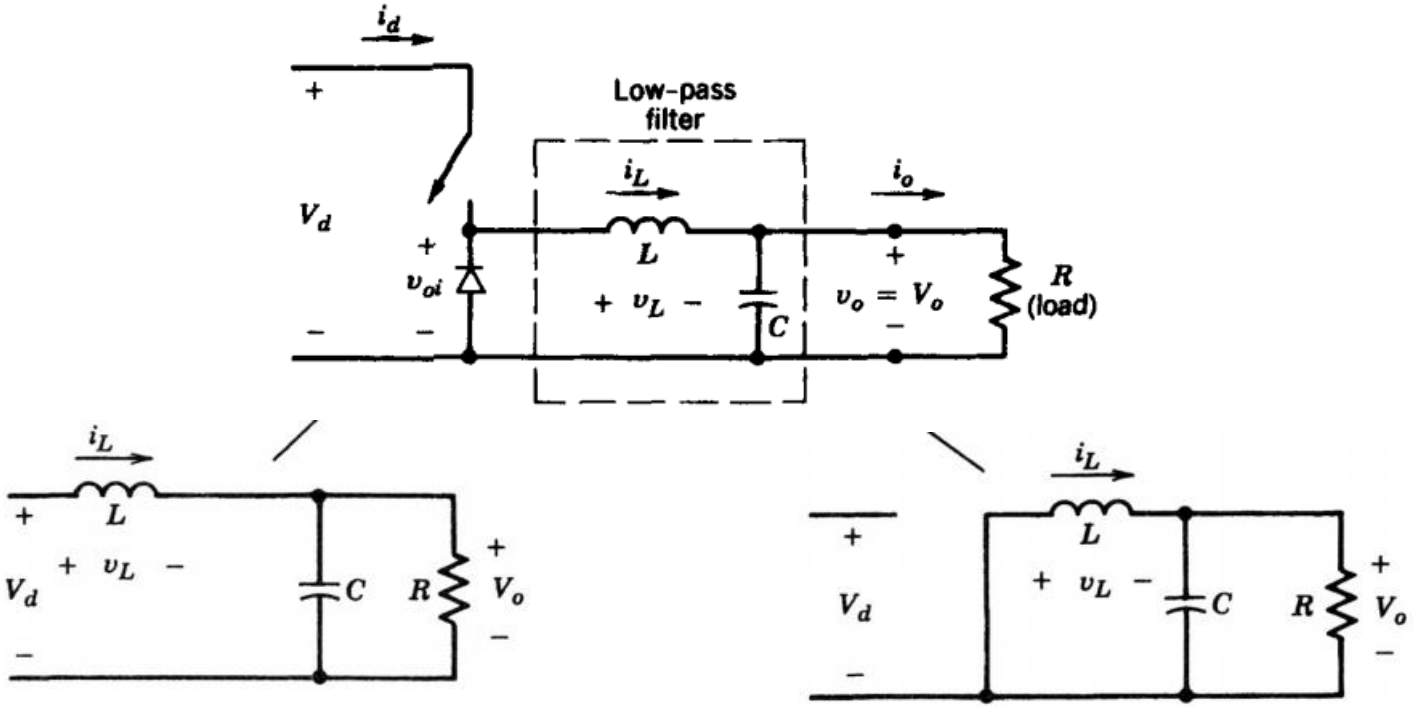
Step-down (Buck) Converter



- Diode is used to overcome the stored inductive energy
- Low-pass filter is used to overcome the fluctuation of the output voltage
- The output voltage v_{oi} (before filtering) consists of a dc component V_0 and the harmonics at the switching frequency f_s and its multiplies
- The cut-off frequency of the filter is selected to be much lower than the switching frequency

Step-down (Buck) Converter

Continuous conduction mode



Switch is ON

- Diode is reverse biased,
- $v_L = V_d - V_0$

Switch is OFF

- Diode is forward biased, i_L continues to flow through diode
- $v_L = -V_0$

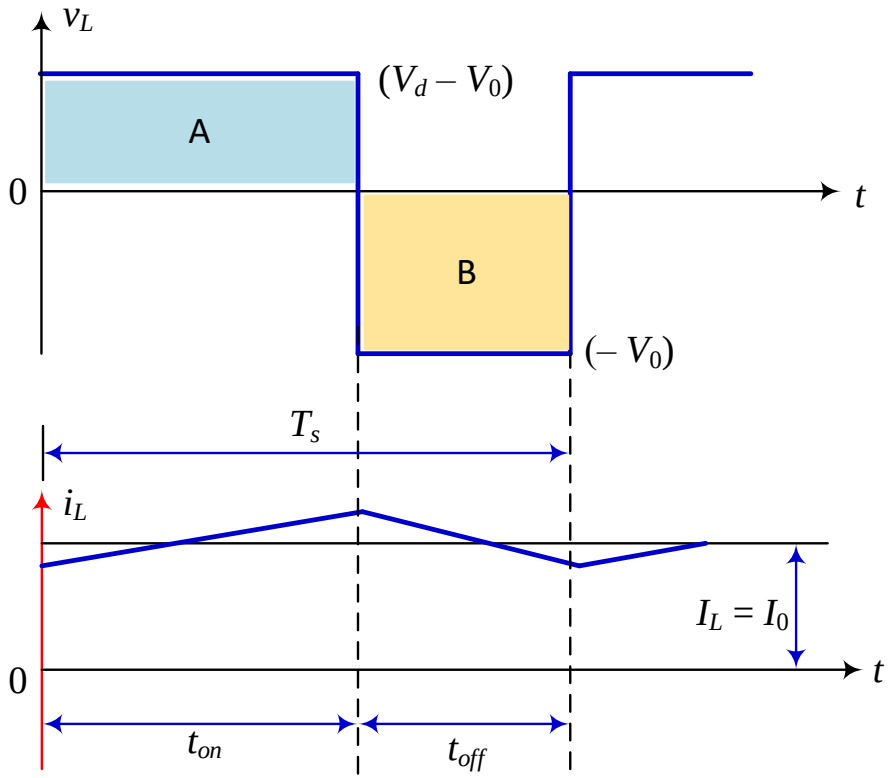
Step-down (Buck) Converter

Continuous conduction mode

$$\int_0^{T_s} v_L dt = \int_0^{t_{on}} v_L dt + \int_{t_{on}}^{t_{off}} v_L dt = 0$$

$$(V_d - V_0)t_{on} = V_0(T_s - t_{0n})$$

$$\Rightarrow \frac{V_0}{V_d} = \frac{t_{on}}{T_s} = D$$



Neglecting power losses associated with all the circuit elements

$$P_d = P_0$$
$$V_d I_d = V_0 I_0$$
$$\Rightarrow \frac{I_0}{I_d} = \frac{V_0}{V_d} = D$$

Step-down (Buck) Converter

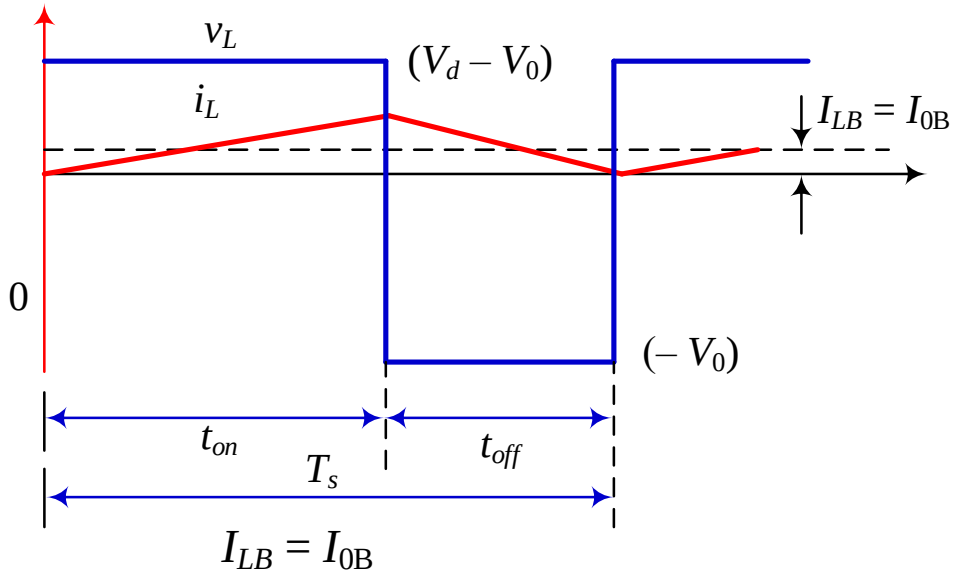
Boundary continuous-discontinuous mode

- The average inductor current (at the boundary)

$$I_{LB} = \frac{i_{L,peak}}{2}$$

- The inductor voltage

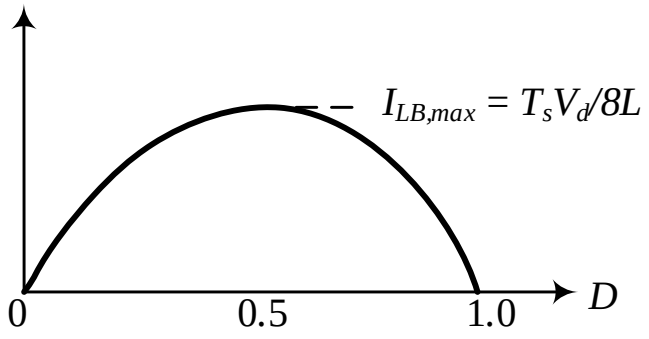
$$v_L = V_d - V_0 = L\frac{di_L}{dt} = L\frac{i_{L,peak} - 0}{t_{on}}$$



$$\Rightarrow I_{LB} = \frac{t_{on}}{2L}(V_d - V_0) = \frac{DT_s}{2L}(V_d - V_0) = I_{0B}$$

- Note that: $D = \frac{V_0}{V_d}$

$$\Rightarrow I_{LB} = I_{0B} = \frac{V_d T_s}{2L} D(1-D) \quad (*)$$



For given values of V_d, T_s, L, and D, if the average output current less than I_{LB} given by equation (*), then i_L will become discontinuous

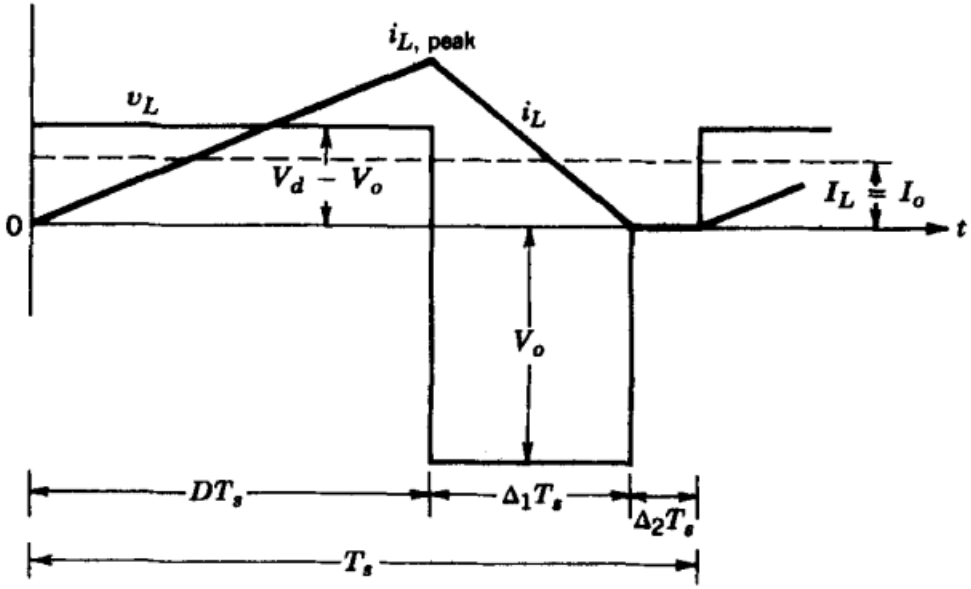
Step-down (Buck) Converter

Discontinuous conduction mode

- Since the average inductor voltage is zero

$$(V_d - V_0)DT_s + (-V_0)\Delta_1 T_s = 0$$

$$\Rightarrow \frac{V_0}{V_d} = \frac{D}{D+\Delta_1} \qquad (1)$$



- Relation between Δ_1 and I_0?

$$I_0 = i_{L,peak}\frac{D+\Delta_1}{2}$$

where $i_{L,peak} = \dfrac{V_0\Delta_1 T_s}{L}$ $\Rightarrow I_0 = \dfrac{V_0\Delta_1 T_s}{L}\dfrac{(D+\Delta_1)}{2}$ (2)

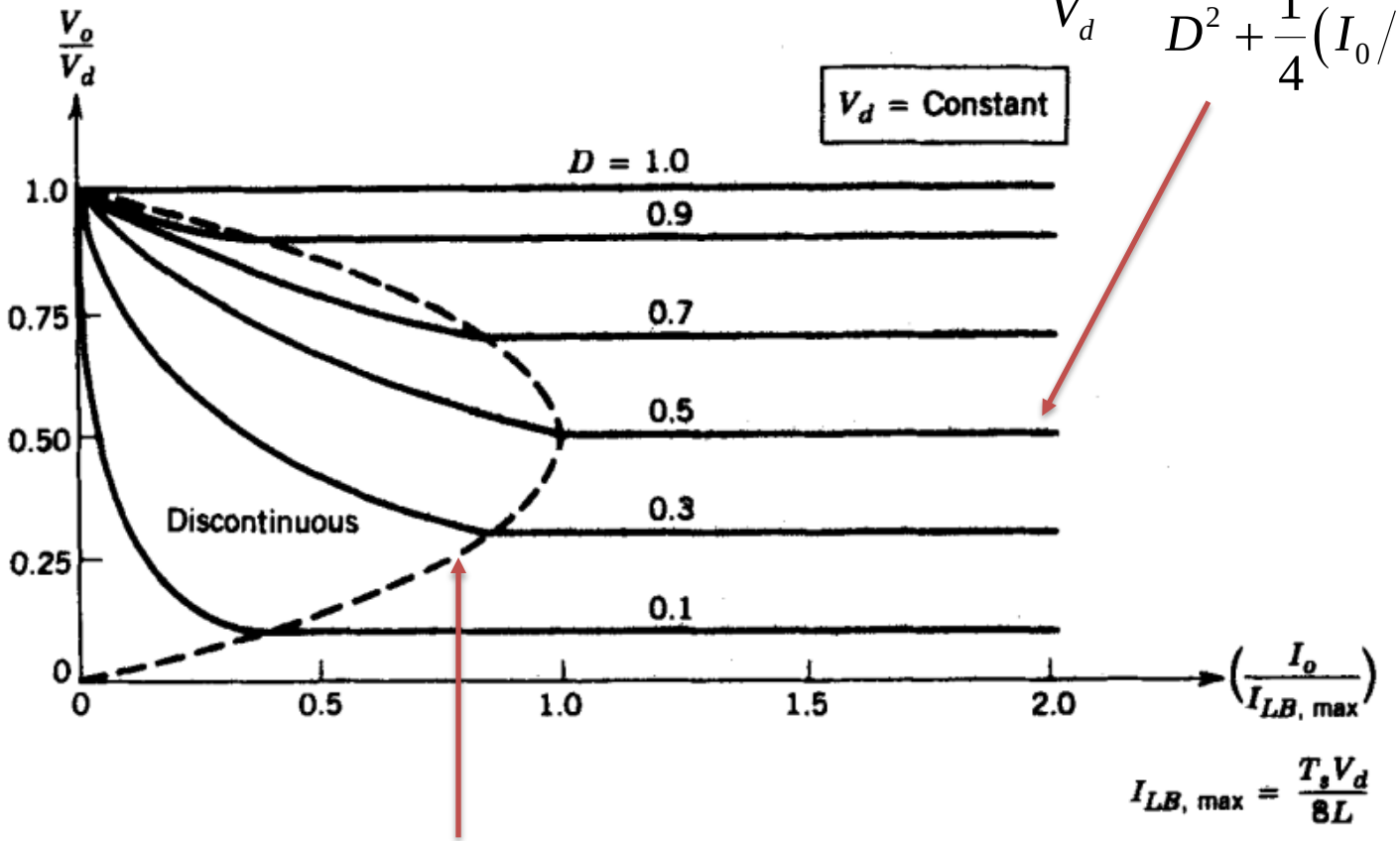
From (1) and (2)

$$\frac{V_0}{V_d} = \frac{D^2}{D^2 + \dfrac{2L}{T_s V_d} I_0} \quad \text{or} \quad \frac{V_0}{V_d} = \frac{D^2}{D^2 + \dfrac{1}{4}\left(I_0 / I_{LB\max}\right)}$$

Step-down (Buck) Converter

Discontinuous conduction mode

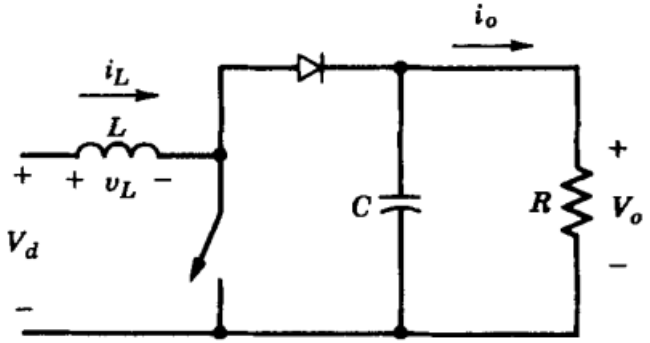
$$\frac{V_0}{V_d} = \frac{D^2}{D^2 + \frac{1}{4}\left(I_0 / I_{LB\max}\right)}$$

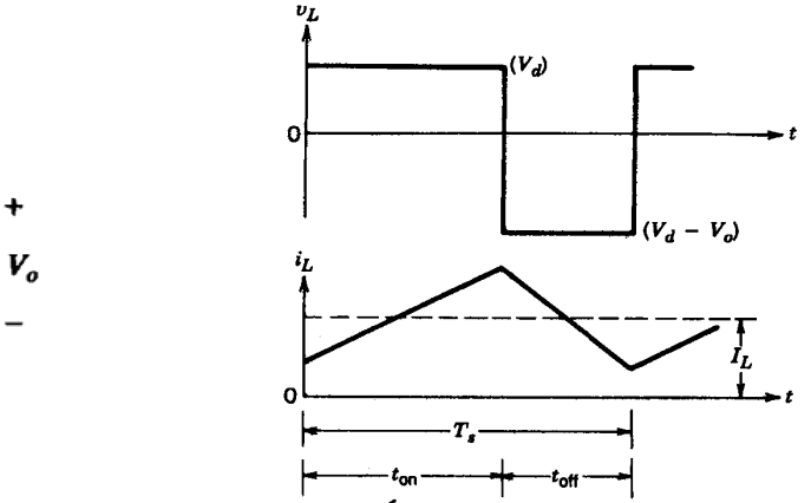


$$\frac{I_{LB}}{I_{LB,\max}} = 4D(1-D) \quad \text{boundary}$$

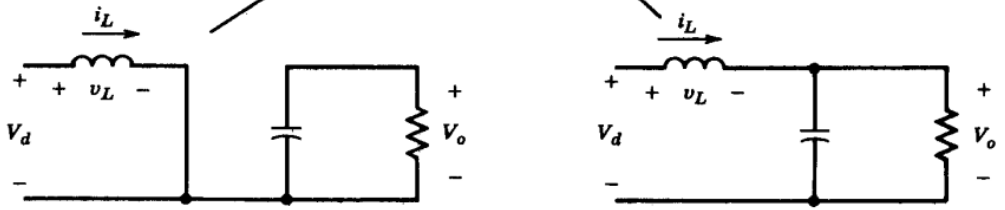
Step-up (Boost) Converter

- A step-up converter produces a higher average output voltage than the dc input voltage





- Switch is ON: diode is reverse biased, inductor stores energy
- Switch is OFF: the output stage receives energy from inductor as well as from the input



$$v_{L,avg} = 0 \Rightarrow \quad V_d t_{on} + (V_d - V_0) t_{off} = 0$$

$$\Rightarrow \frac{V_0}{V_d} = \frac{1}{1-D}$$

Step-up (Boost) Converter

Boundary continuous-discontinuous mode

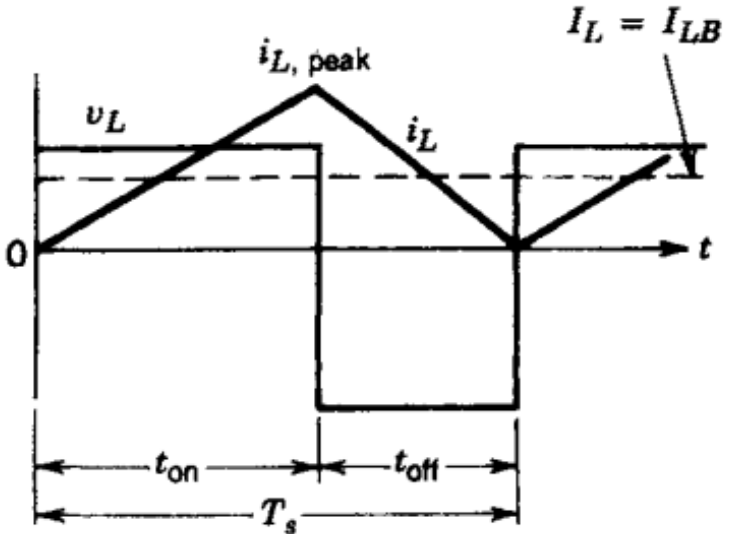
- The average value of the inductor curren

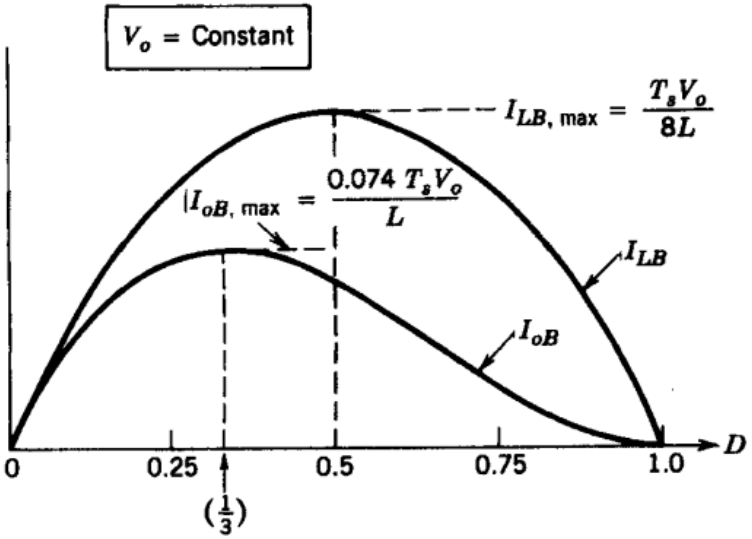
$$I_{LB} = \frac{1}{2} i_{L,peak} = \frac{1}{2} \frac{V_d}{L} t_{on} = \frac{T_s V_0}{2L} D(1-D)$$

- The average output current

$$I_{0B} = \frac{T_s V_0}{2L} D(1-D)^2$$

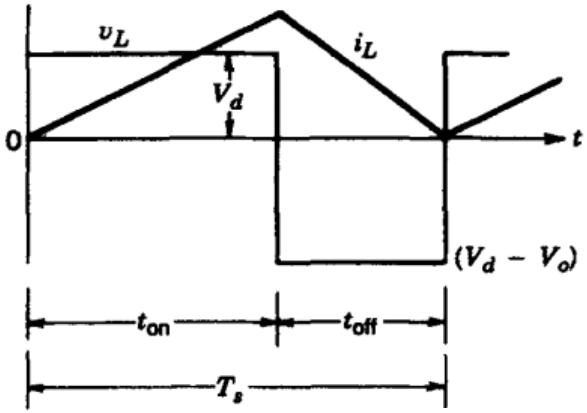
With a constant V_0, if the average load current drops below I_{0B} (and, hence, the average inductor current below I_{LB}), the conduction will become discontinous

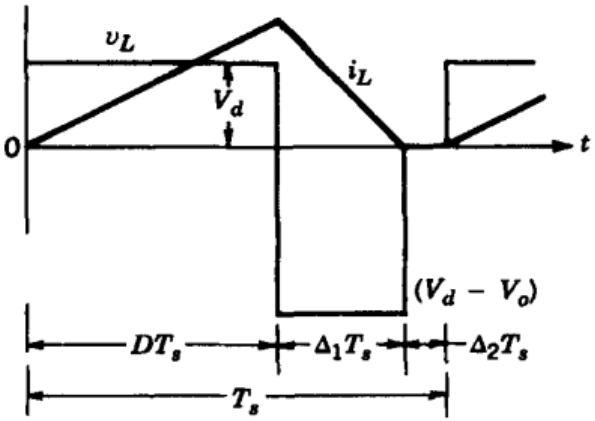


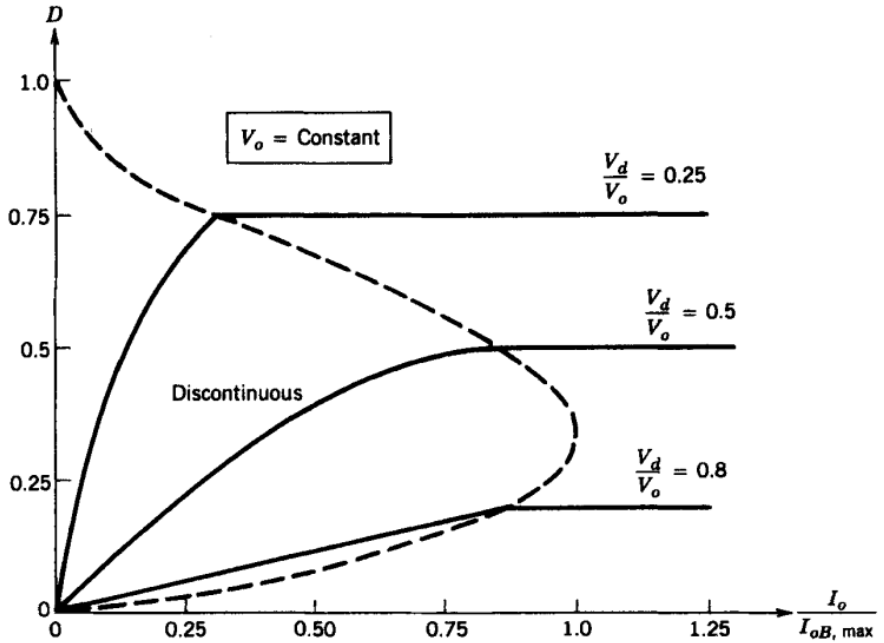


Step-up (Boost) Converter

Discontinuous conduction mode







$$I_{oB,\,max} = 0.074\,\frac{T_s V_o}{L}$$

Step-up (Boost) Converter

Homework:

STEP-DOWN CONVERTERS

7-1 In a step-down converter, consider all components to be ideal. Let $v_o \simeq V_o$ be held constant at 5 V by controlling the switch duty ratio D. Calculate the minimum inductance L required to keep the converter operation in a continuous-conduction mode under all conditions if V_d is 10–40 V, $P_o \geq$ 5 W, and $f_s = 50$ kHz.

7-2 Consider all components to be ideal. Assume $V_o = 5$ V, $f_s = 20$ kHz, $L = 1$ mH, and $C = 470$ μF. Calculate ΔV_o (peak–peak) if $V_d = 12.6$ V, and $I_o = 200$ mA.

STEP-UP CONVERTERS

7-7 In a step-up converter, consider all components to be ideal. Let V_d be 8–16 V, $V_o = 24$ V (regulated), $f_s = 20$ kHz, and $C = 470$ μF. Calculate L_{min} that will keep the converter operating in a continuous-conduction mode if $P_o \geq 5$ W.

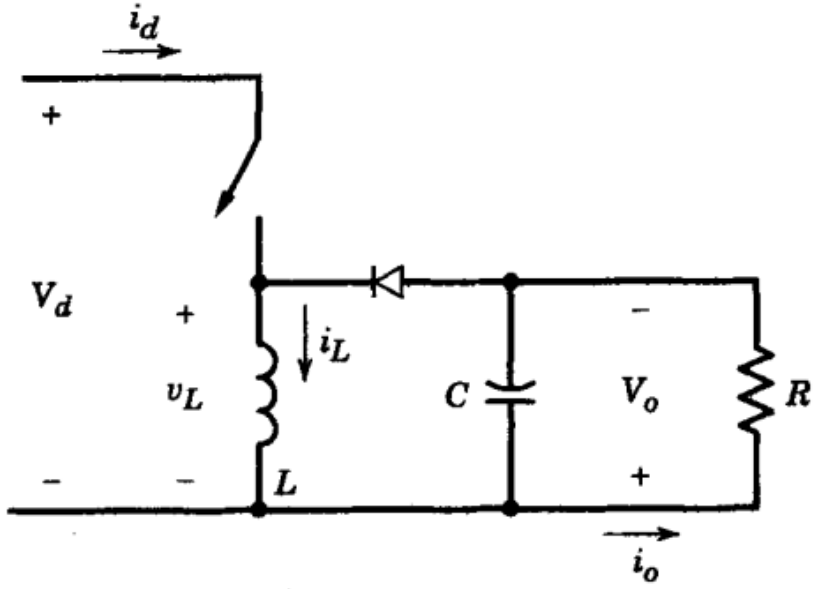
7-8 In a step-up converter, $V_d = 12$ V, $V_o = 24$ V, $I_o = 0.5$ A, $L = 150$ μH, $C = 470$ μF, and $f_s =$ 20 kHz. Calculate ΔV_o (peak–peak).

Buck-Boost Converter

- Buck-Boost converter can be obtained by the casecade connection of the two basic converters: step-down converter and step-up converter
- The output – to – input voltage conversion ratio is the product of of the conversion ratios of the two converters in casecade

$$\frac{V_0}{V_d} = D\frac{1}{1-D}$$

- Switch is ON: diode is reverse based, inductor receives energy from the input
- Switch is OFF: energy stored in the inductor is transferred to the output



Buck-Boost Converter

Continuous conduction mode

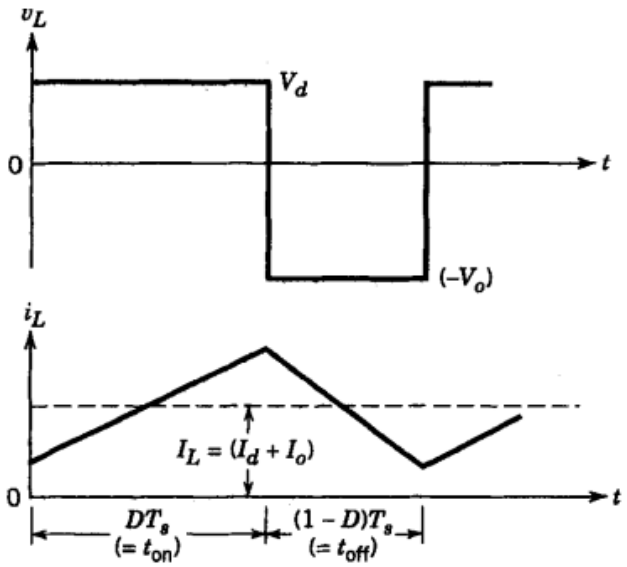
Since $v_{L,av} = 0$

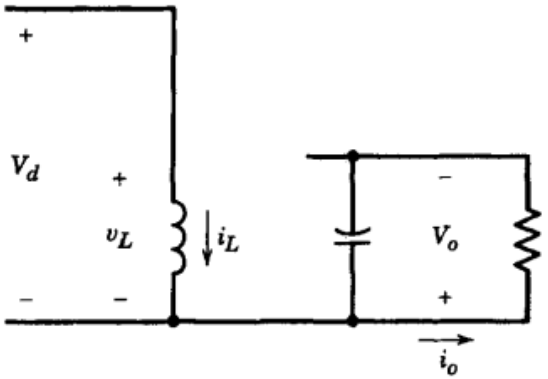
$$V_d DT_s + (-V_0)(1-D)T_s = 0$$

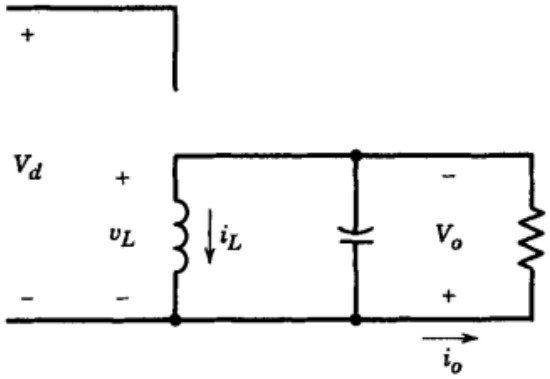
$$\Rightarrow \frac{V_0}{V_d} = \frac{D}{1-D}$$

Assume $P_d = P_o$

$$\frac{I_0}{I_d} = \frac{1-D}{D}$$







Buck-Boost Converter

Boundary between continuous and discontinuous

i_L goes to zero at the end of off interval

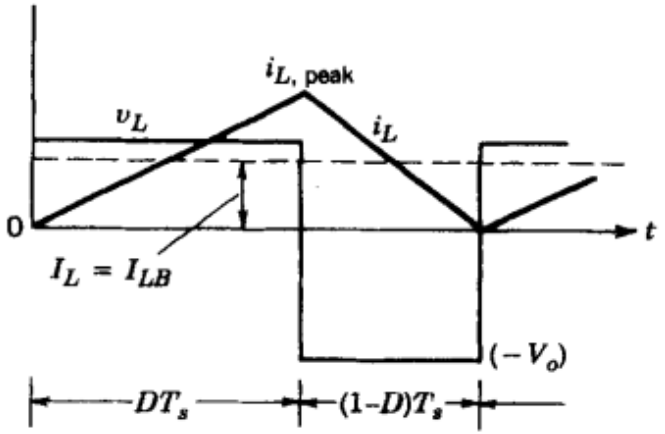
$$i_{LB} = \frac{1}{2} i_{L,peak} = \frac{T_s V_d}{2L} D$$

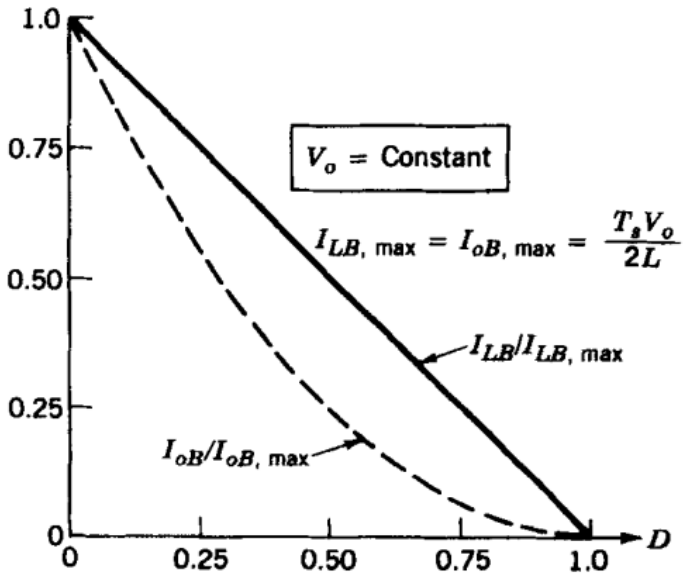
Since the average capacitor current is zero

$$I_0 = I_L - I_d$$

$$\Rightarrow I_{LB} = \frac{T_s V_0}{2L}(1-D)$$

$$\Rightarrow I_{oB} = \frac{T_s V_0}{2L}(1-D)^2$$





Buck-Boost Converter

Discontinuous conduction mode

Since $v_{L,av} = 0$

$$V_d DT_s + (-V_0)\Delta_1 T_s = 0$$

$$\Rightarrow \frac{V_0}{V_d} = \frac{D}{\Delta_1}$$

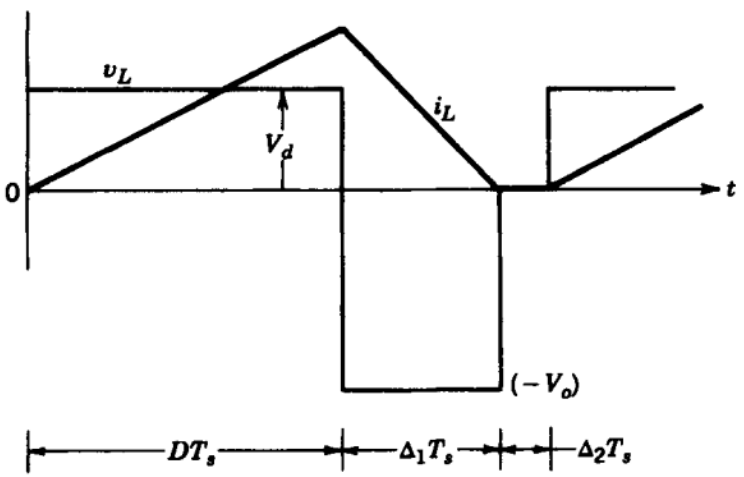
$$\Rightarrow \frac{I_0}{I_d} = \frac{\Delta_1}{D}$$

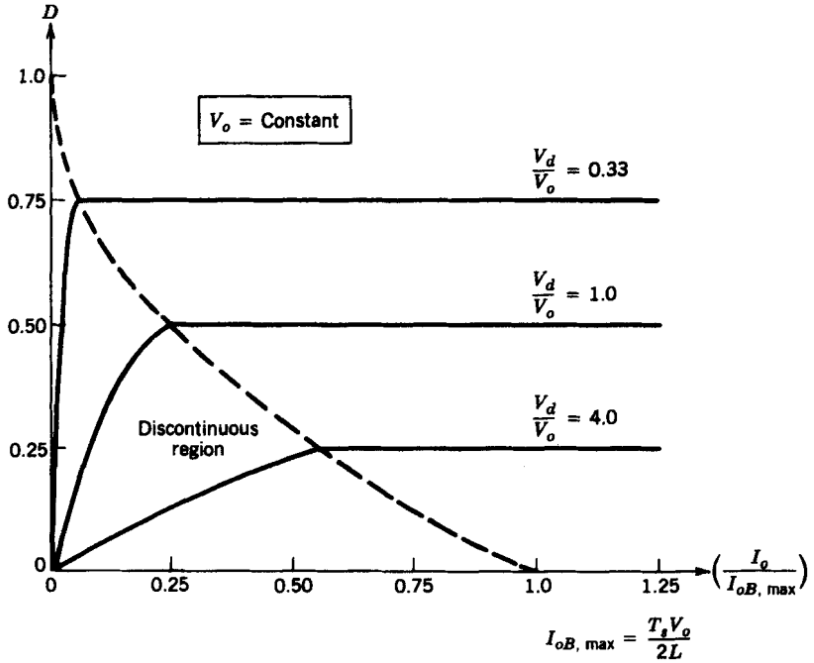
The average inductior current

$$I_L = \frac{V_d}{2L} DT_s \left(D + \Delta_1\right)$$

and

$$D = \frac{V_0}{V_d}\sqrt{\frac{I_o}{I_{oB,\max}}}$$





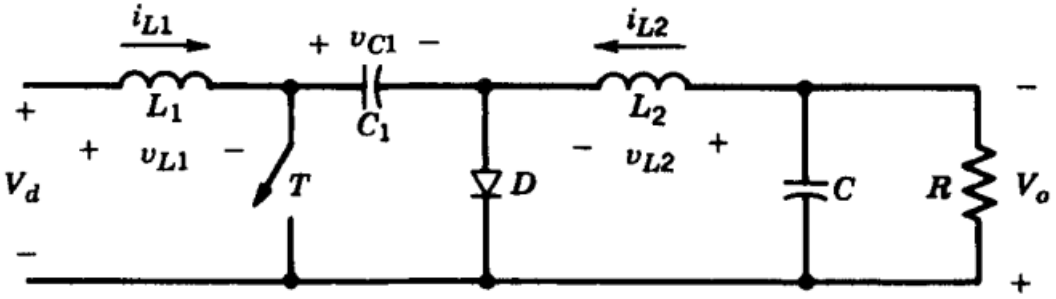
Buck-Boost Converter

Ex: A buck-boost converter operating at 20 kHz, L = 0.05 mH, C is sufficiently large, V_d = 15V. The output to be regulated at 10V, load power is 10W. Calculate duty ratio D.

Cúk Converter

- Cúk converter is a type of buck-boost converter
- In the steady-state

$$V_{C1} = V_d + V_0$$



Switch is OFF:

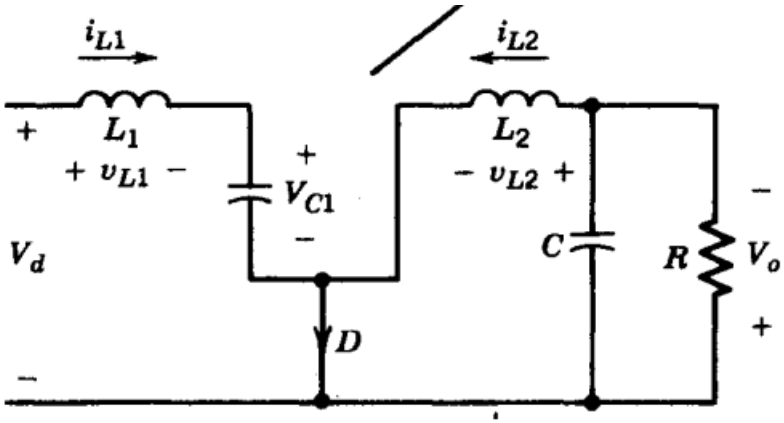
- i_{L1}, i_{L2} flow through the diode
- Capacitor C_1 is charged by the energy from both the input and L_1
- i_{L1} is decreased because $V_{C1} > V_d$
- Stored energy in L_2 transfers to the output, i_{L2} is decreased

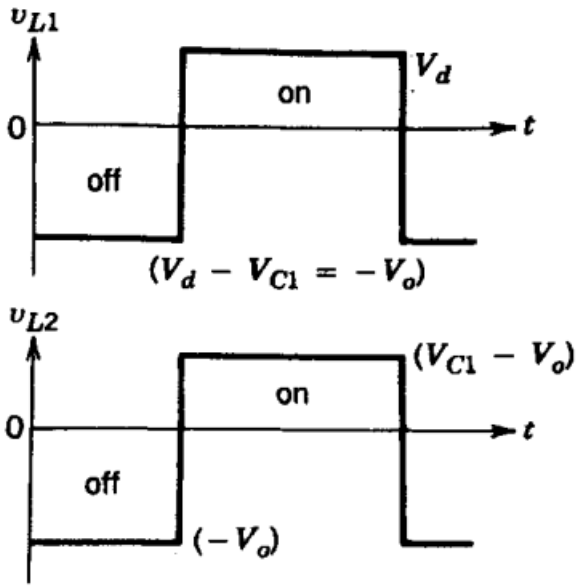
Switch is ON:

- V_{C1} reverse biases the diode
- i_{L1}, i_{L2} flow through the switch
- Since $V_{C1} > V_0$, C_1 is discharged, transferring energy to the output and L_2, i_{L2} is increased
- L_1 receives energy from the input, iL_1 is increased

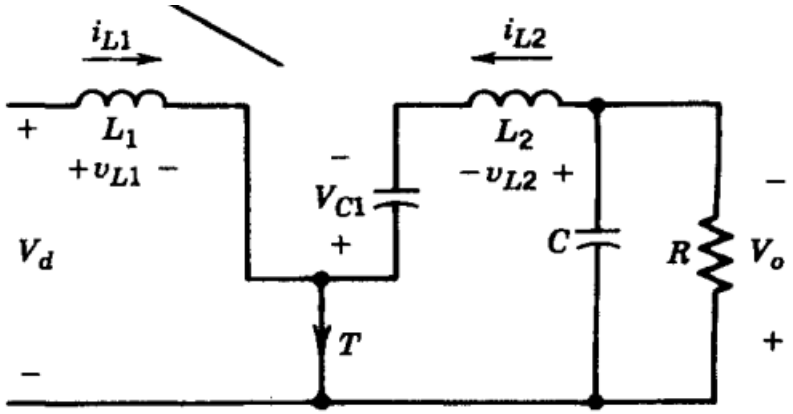
Cúk Converter

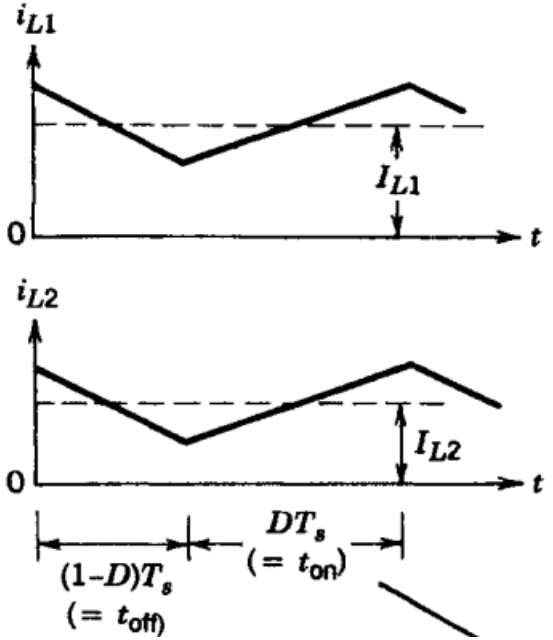
Switch is OFF:





Switch is ON:





Cúk Converter

- Assume that i_{L1} and i_{L2} are continuous
- Since $v_{L1,av} = 0$

$$V_d DT_s + (V_d - V_{C1})(1-D)T_s = 0$$

$$\Rightarrow V_{C1} = \frac{1}{1-D} V_d$$

- Since $v_{L2,av} = 0$

$$(V_{C1} - V_0) DT_s + (-V_0)(1-D)T_s = 0$$

$$\Rightarrow V_{C1} = \frac{1}{D} V_0$$

- Finally

$$\frac{V_0}{V_d} = \frac{D}{1-D} \qquad \frac{I_0}{I_d} = \frac{1-D}{D}$$

where $I_{L1} = I_d; \quad I_{L2} = I_0$

